DockForgeAI

Reproducible Molecular Docking, Virtual Screening and Hit Prioritization Report

Project version 1.0.0
Target: 1HSG Reference ligand: MK1
Generated: 2026-08-01 10:05 UTC

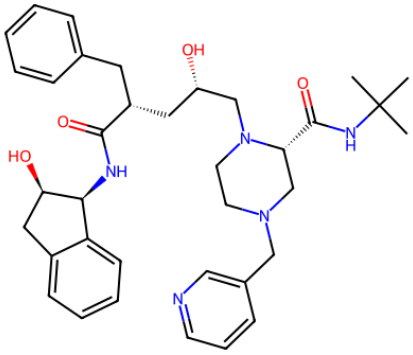
Software-generated scientific report. Docking scores are approximate ranking values, not experimental binding free energies.

Executive summary

DockForgeAI evaluated 1HSG with reference ligand MK1. The validated top pose reported a Vina score of -11.575 kcal/mol and crystallographic heavy-atom RMSD of 0.518 Å (Validated strongly).

Interaction analysis recorded 141 geometric contacts, including 54 contacts involving catalytic Asp25. The demonstration screen processed 3 compounds with 3 successful runs.

Best demonstration compound: 3, score -2.848 kcal/mol. This is a software demonstration, not evidence of a validated inhibitor; experimental validation is required.



MK1_reference | crystallographic reference

1. Project and computational environment

Software	Version
python	3.12.13
rdkit	2025.9.5
vina	AutoDock Vina v1.2.7
meeko	0.7.1
gemmi	0.7.5
streamlit	1.60.0
py3Dmol	2.5.3
git_commit	663cb37

2. Protein structure and preparation

Field	Value
PDB ID	1HSG
Title	CRYSTAL STRUCTURE AT 1.9 ANGSTROMS RESOLUTION OF HUMAN IMMUNODEFICIENCY VIRUS (HIV) II PROTEASE COMPLEXED WITH L-735,524, AN ORALLY BIOAVAILABLE INHIBITOR OF THE HIV PROTEASES
Method	X-RAY DIFFRACTION
Resolution	2.0 Å
Chains	A, B
Residues	198
Waters removed	127

3. Ligand preparation

The crystallographic MK1 ligand was extracted, chemistry-corrected, hydrogenated and prepared for redocking using the existing validated workflow.

4. Docking box and receptor assumptions

Input PDB did not encode explicit protonation states for the catalytic Asp25 dyad; the preparation workflow used Meeko's default residue templates and assumed standard carboxylate-type Asp side chains for both catalytic residues.

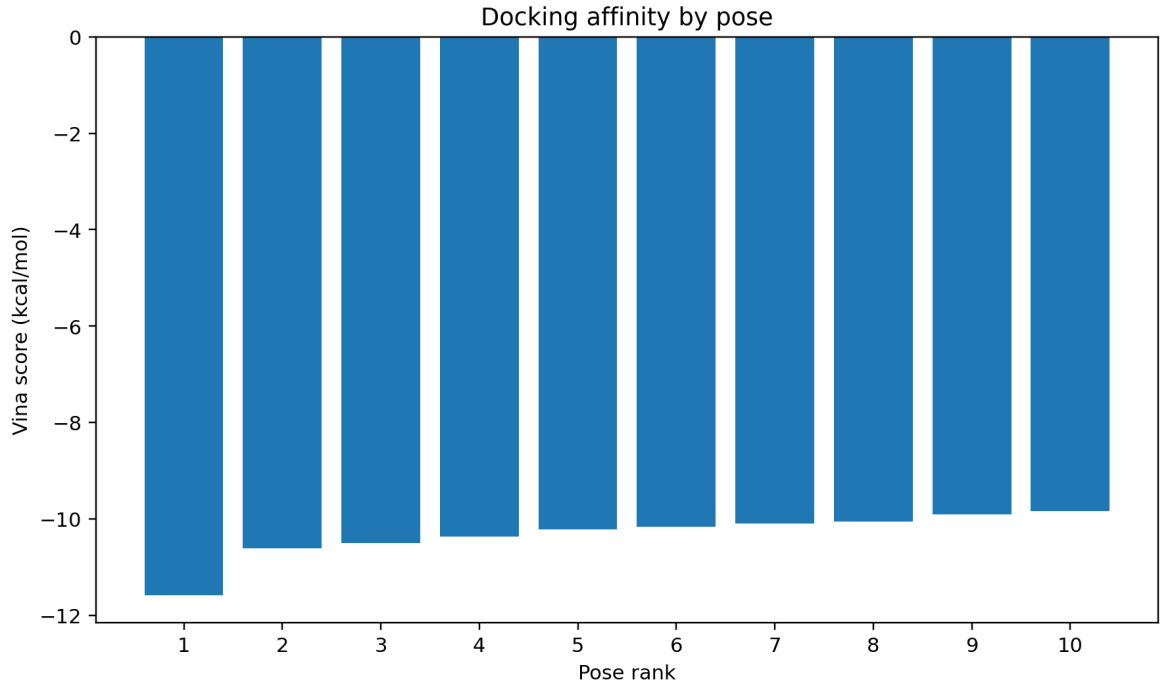
Docking box: {'ligand_sdf': 'data/processed/crystallographic_ligands/1HSG_MK1_crystallographic_corrected.sdf', 'padding_angstrom': 5.0, 'bounds': {'min_x': 4.207, 'max_x': 19.061, 'min_y': 17.363, 'max_y': 27.577, 'min_z': 1.541, 'max_z': 10.176}, 'center': {'x': 11.634, 'y': 22.47, 'z': 5.859}, 'size': {'x': 24.854, 'y': 20.214, 'z': 20.0}, 'span': {'x': 14.854, 'y': 10.214, 'z': 8.635}, 'warnings': []}

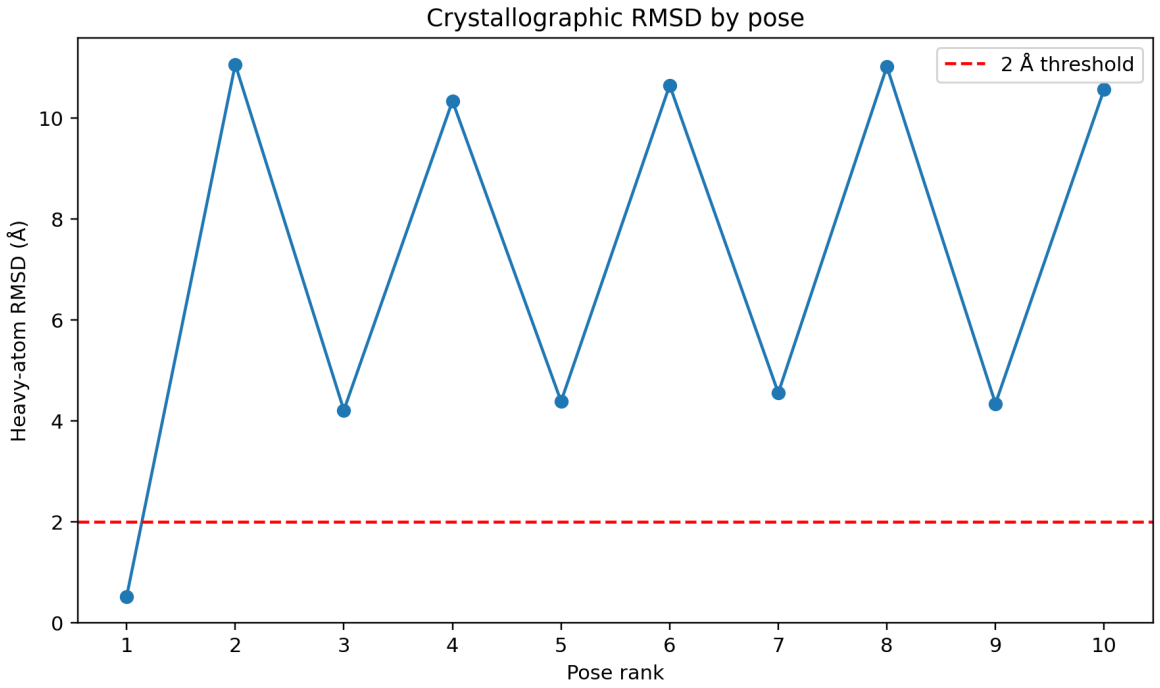
5. Redocking validation

Best score: -11.575 kcal/mol; minimum RMSD: 0.518 Å; interpretation: Validated strongly.

6. Pose ranking table

Rank	Vina score (kcal/mol)	Crystal RMSD (Å)	Pass ≤2 Å
1	-11.575	0.518	True
2	-10.601	11.061	False
3	-10.497	4.207	False
4	-10.37	10.334	False
5	-10.21	4.382	False
6	-10.159	10.647	False
7	-10.089	4.55	False
8	-10.054	11.016	False
9	-9.91	4.338	False
10	-9.832	10.565	False



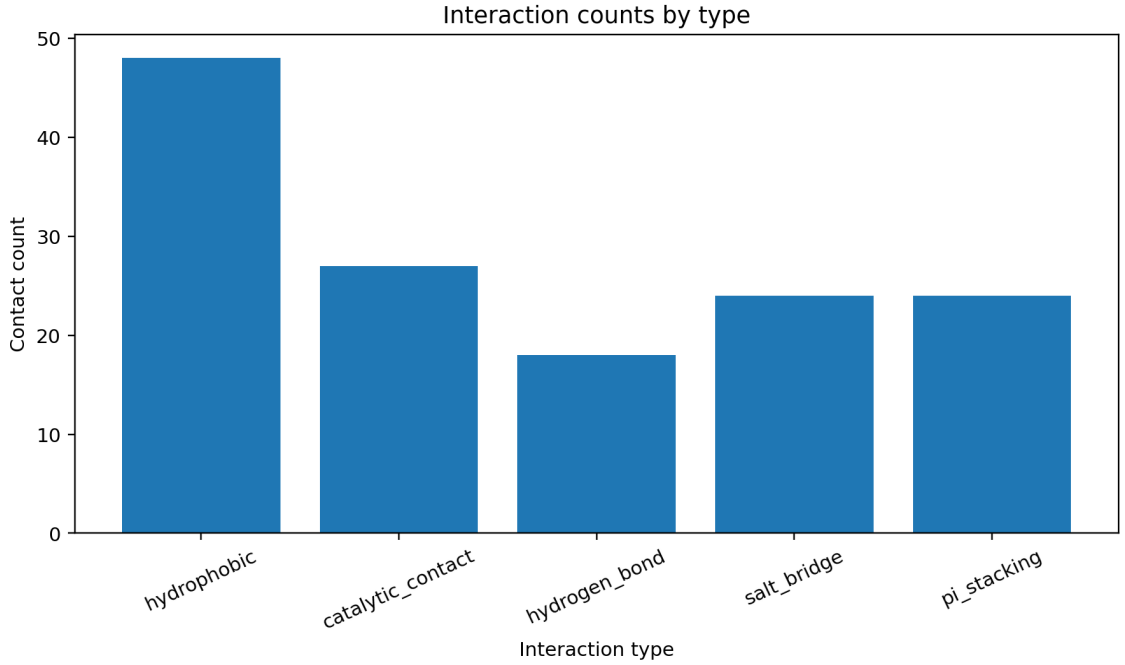

Crystallographic RMSD by pose
2 Å threshold
Heavy-atom RMSD (Å)
Pose rank
0
2
4
6
8
10
1
2
3
4
5
6
7
8
9
10

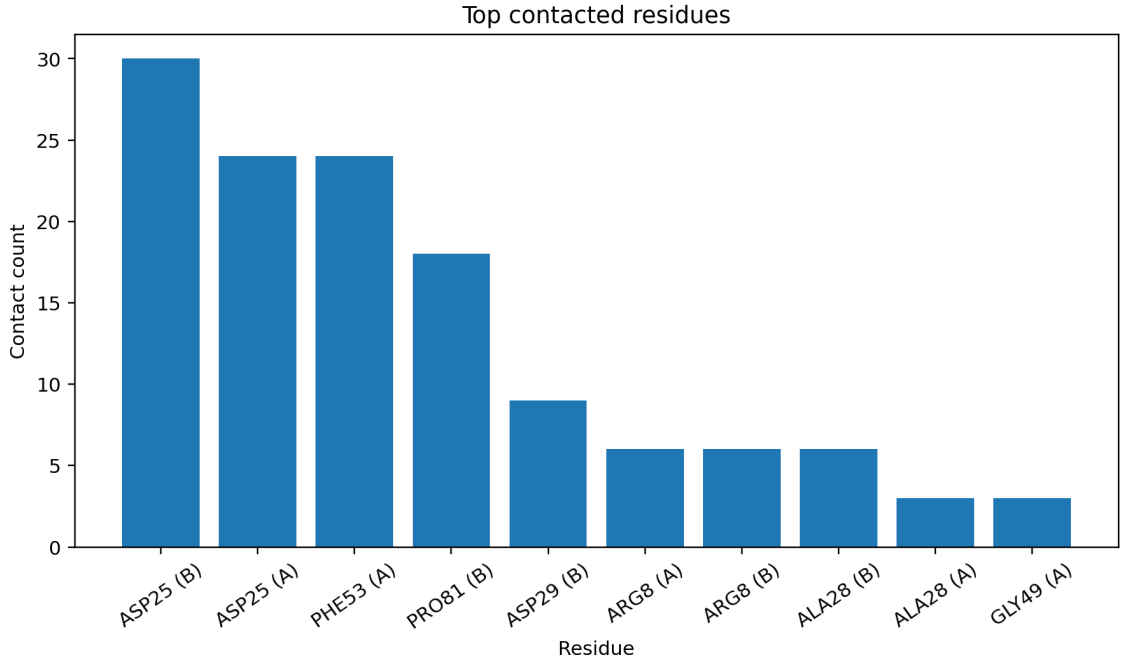
7. Interaction analysis

Interaction type	Count
hydrophobic	48
catalytic_contact	27
hydrogen_bond	18
salt_bridge	24
pi_stacking	24

8. Catalytic Asp25 findings

The geometric analysis recorded 54 contacts involving ASP25 in chains A or B. Contact predictions depend on the stated geometric fallback assumptions.





9. Virtual-screening demonstration

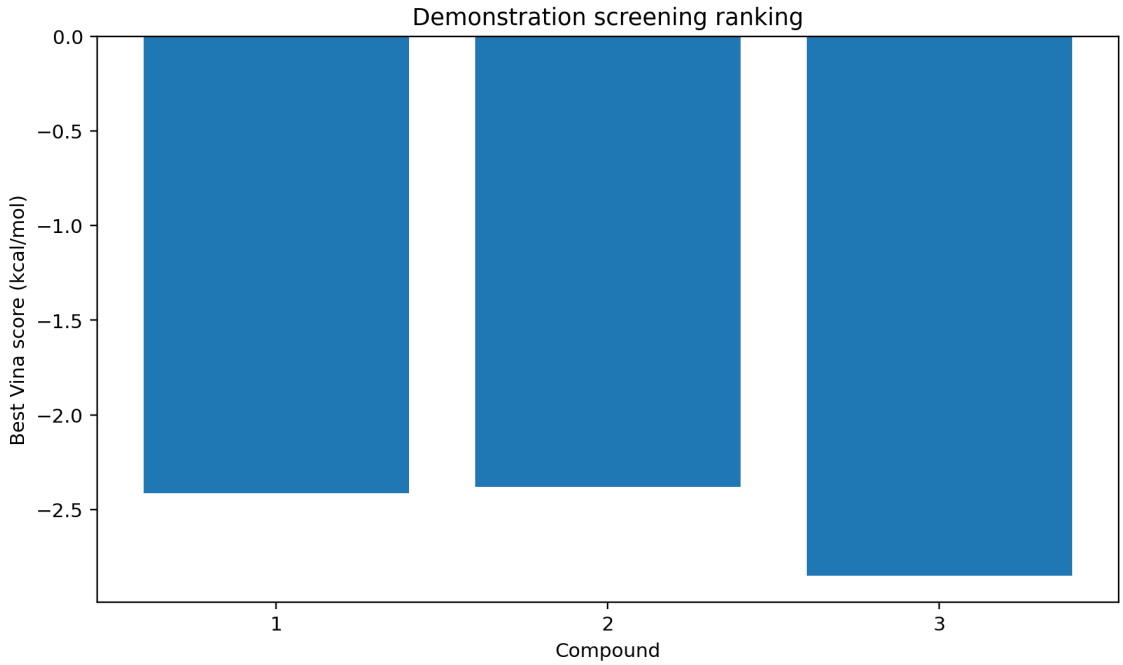
This three-compound run validates the software path only. It is not a biological discovery study and the compounds are not validated inhibitors.

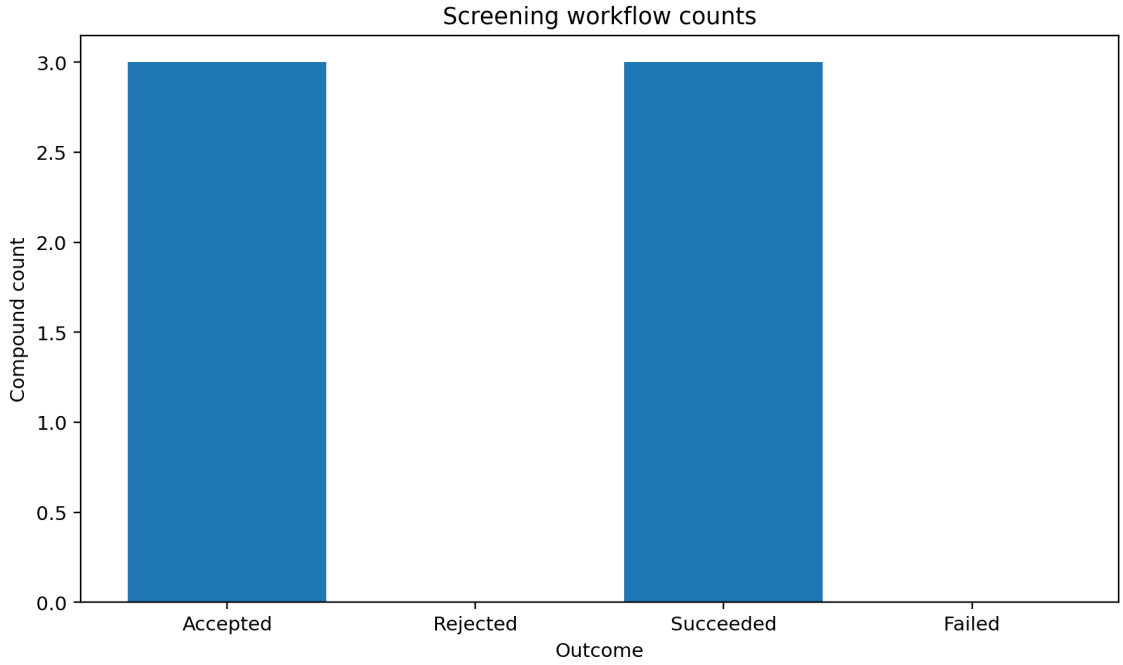
10. Screening ranking

Compound	Best Vina score	Status	Runtime (s)
1	-2.412	success	0.74
2	-2.381	success	0.56
3	-2.848	success	0.557

11. Drug-likeness and structural-alert profile

Reported RDKit descriptors and structural alerts support prioritization; they are not full ADMET predictions.





12. Hit prioritization

Compound	Rank	Consensus	Vina score	QED	Alerts
demo	1	1.0	-6.0	0.8	0.0
demo	2	0.65	-6.0	0.407	0.0

External ADMET provenance

Not supplied; RDKit descriptors are not full ADMET predictions.

13. Scientific limitations

- Docking scores are approximate ranking values, not experimental binding free energies.
- The three-compound screen demonstrates software operation and does not identify validated inhibitors.
- RDKit descriptors are not full ADMET predictions.
- All computational predictions require experimental validation.

14. Reproducibility and file paths

Source inputs (project-relative):

```
data/processed/protein_reports/1HSG_inspection.json
data/processed/docking_results/1HSG_MK1_redocking_summary.json
data/processed/interaction_reports/interactions.csv
data/processed/screening_tables/screening_scores.csv
data/processed/prioritization_reports/consensus_ranking.csv
```

15. Conclusion

The validated 1HSG/MK1 redocking result reproduces the crystallographic pose strongly (0.518 Å). Downstream screening and prioritization remain computational demonstrations requiring experimental confirmation.